Justin Le

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Interests

Machine learning (generative models) and stochastic optimization

Employment History

2025 – Present	**Teaching Assistant**, University of Texas at Austin
2024 – 2024	**Instructional Assistant / Grader**, Arizona State University
2022 – 2023	**Instructor,** Mathnasium

Education

2025 – 2030	**Ph.D, University of Texas at Austin**, Mathematics **GPA: 4.0/4.0**
2024 – 2025	**M.A, Arizona State University**, Mathematics **GPA: 4.0/4.0**
2021 – 2024	**B.S, Arizona State University**, Mathematics (Honors) Thesis title: *Diffusion Models to Alleviate Class Imbalance* **GPA: 4.0/4.0**

Publications

1. J. Le, "Generative modeling with diffusion," *SIAM Undergraduate Research Online*, vol. 18, pp. 213–229, Jun. 2025. DOI: 10.1137/24S1717993.
2. F. Cao, K. Johnston, T. Laurent, J. Le, and S. Motsch, *Generative diffusion models from a pde perspective*, **submitted**. arXiv: 2501.17054.

Talks and Presentations

April 2026	"Generative Diffusion Models from a PDE Perspective" UT Austin Junior Analysis Seminar
January 2026	"Tweedie's Formula and Score-Based Generative Diffusion Models" UT Austin Applied Math Reading Group
October 2025	"Neural Operators for Learning Mappings Between Function Spaces" UT Austin Junior Analysis Seminar
November 2024	"Diffusion Models to Alleviate Class Imbalance" ASU Undergraduate Honors Thesis Defense

Research Experience

2026 **SIAM Mathematical Problems in Industry Workshop**, Drexel University

- Worked with a representative from RTX Corporation to improve a Kalman filter-type algorithm with applications to deep learning.
- Constructed and applied alternative approximations to the covariance matrix of a Kalman filter.

SIAM Graduate Student Mathematical Modeling Camp, University of Delaware

- Studied a kinetic Monte-Carlo model of disease spread and cell-listing algorithms to improve computational time for large population sizes.
- Derived heuristics for cell length and the duration for which a cell-list can be maintained.

2022 – 2025 **Undergraduate Research Assistant**, Arizona State University
Advisor: Dr. Sebastien Motsch

- **First project**: Designed, trained, and evaluated convolutional neural networks for semantic segmentation on a dataset of slime mold laboratory images. Computed the geometry of slime mold samples with the results from segmentation.
- **Second project**: Designed, trained, and evaluated diffusion models for synthesizing data. This synthetic data was then applied to a dataset of credit card transactions to improve a classifier's detection of credit card fraud.

Teaching Experience

2025 – Present **Teaching Assistant**, University of Texas at Austin

Multivariable Calculus (two sections)	Fall 2026
Real Analysis I	Spring 2026
Real Analysis II	Spring 2026
Integral Calculus (two sections)	Fall 2025

2024 – 2024 **Instructional Assistant**, Arizona State University

Mathematics for Business Analysis	Fall 2024
Calculus for Engineers II (two sections)	Summer 2024
Mathematics for Business Analysis (two sections)	Summer 2024
Discrete Mathematical Structures	Spring 2024

Awards

2026 **Frank Gerth III Teaching Assistant Award (University of Texas at Austin)** – Awarded to graduate teaching assistants to recognize excellence in mathematics education.

2024 **Dean's Medal (Arizona State University)** – Awarded to one graduating student in the mathematics/statistics department each semester to recognize academic achievement.

Moeur Award (Arizona State University) – Awarded to graduating students who maintain a 4.0 GPA throughout their undergrad.

Outreach and Service

2025-2026 Organized the Sophex seminar at UT Austin. This is a weekly seminar that allows first-year math PhD students to practice giving talks.

Outreach and Service (continued)

2025-26 Mentored for the Directed Reading Program at UT Austin. I mentored three students on statistical inference across three semesters, including an online version of the Directed Reading Program in Summer 2026.

Skills

Programming	Python, C/C++, Java, MATLAB, SQL, LaTeX
Data Science	PyTorch, Matplotlib, Pandas, MySQL
Data Processing	Apache Spark, Apache Hadoop, Slurm Scripting
Computer	Bash (Linux), Git, PyCharm, VSCode